

Mathanraj Rajendran

Research Assistant
Smart Construction & Systems Laboratory
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EDUCATION

Master of Science in Global Smart City 2023-2025
Sungkyunkwan University, South Korea

- Thesis Title: Dimensional quality assessment of structural steel members using terrestrial laser scan data
- Advisors: Professor Sung-Han Sim, and Professor Min-Koo Kim
- GPA: 4.31/4.5

Bachelor of Engineering in Civil Engineering 2014-2018
Anna University, India

- GPA: 7.3/10.0, Graduated in First Class

RESEARCH INTERESTS

- | | |
|---------------------------------|--|
| 1. Construction robotics | 2. Computer vision |
| 3. Scan-to-BIM and digital twin | 4. Automated inspection and monitoring |

PUBLICATIONS

Published:

1. Ayembilla, F.B., **Rajendran, M.**, Kim, M. K., Semantic Segmentation of MEP Point Cloud Scenes Using Point Cloud Completion Approach, ICCEPM 2025, Jeju. 2026
<https://doi.org/10.6106/ICCEPM.2025.0306>
2. Duorinaah, F. X., **Rajendran, M.**, Kim, T. W., Kim, J. I., Lee, S., Lee, S., & Kim, M. K. (2025). Human and Multi-Robot Collaboration in Indoor Environments: A Review of Methods and Application Potential for Indoor Construction Sites. Buildings, 15(15), 2794. <https://doi.org/10.3390/buildings15152794> 2025
3. **Rajendran, M.**, Sim, S. H., Kim, M. K., Choi, Y. K., & Yi, C. Y. (2025). Automated dimensional quality assessment of steel members using TLS and enhanced edge line estimation. Measurement, 118990. <https://doi.org/10.1016/j.measurement.2025.118990> 2025
4. **Rajendran, M.**, Sim, S. H., Kim, M. K., & Choi, Y. K. (2024). Dimensional Quality Assessment of Steel H-Beams Using Terrestrial Laser Scan Data. In International conference on construction engineering and project management (pp. 264-270). Korea Institute of Construction Engineering and Management. 2024
<https://doi.org/10.6106/ICCEPM.2024.0264>

In progress:

5. **Rajendran, M.**, Duorinaah, F. X., & Kim, M. K. Advancing Indoor Construction with Autonomous Robots: Exploring Mobile Robot Applications, Operations, and

Challenges in Building Environments. Journal of Information Technology in Construction. *(Under review)*

6. **Rajendran, M.**, Ayembilla, F.B., Sim, S. H., & Kim, M. K., Automated Shape-Aware Dimensional Classification and Semantic-Instance Segmentation of Steel Structures from Occluded Point Cloud Data. Automation in Construction *(Under review)*

RESEARCH EXPERIENCE

Research Assistant

2026-Present

Smart Construction & Systems Lab, Chungbuk National University
Advisor: Prof. Min-Koo Kim

- Develop framework for semantic segmentation of building indoors using foundation models and Vision Language Models (VLMs)
- Develop methods to improve semantic and instance segmentation accuracy of steel structures using synthetic point cloud data and deep learning-based occlusion reconstruction models.
- Develop synthetic data generation methods and AI-assisted techniques for real-time object detection for bin picking using robotic arms.

Research Assistant

2025-2025

Built Environment Research Center, Sungkyunkwan University
Advisor: Prof. Min-Koo Kim

- Investigated mobile robot technical challenges of robot operation in indoor confined construction environments.
- Studied general human-robot collaboration techniques and analyzed its adaptability and limitations to indoor construction processes.

Graduate Research Assistant

2023-2025

Smart Construction Laboratory, Sungkyunkwan University
Smart Construction & Systems Laboratory, Chungbuk National University
Advisors: Prof. Sung-Han Sim, and Prof. Min-Koo Kim

- Developed an automated method for accurate cross-sectional level detailed dimensional assessment of structural steel using terrestrial laser scanning point cloud data.
- Proposed a precise edge line estimation method to estimate actual edges of target object while using low-resolution TLS point clouds.

PROJECTS

Semantic Segmentation of Indoor Point Clouds

2026-Present

- Developing approaches for improving semantic segmentation accuracy of building by integrating images and point clouds with zero-shot techniques and Vision Language Models (VLMs)

Bin Picking Using Robotic Arm Manipulation

2025-Present

- Developing methods for real-time bin picking using robotic arms in controlled environments, integrating AI, computer vision, and multi-modal sensors.

Dimensional Quality Assessment of Precast Concrete Panels

2023-2025

- Developed a methodology for precise dimensional quality assessment of precast concrete panels by estimating edge lines from point cloud data.

- Proposed a data-fusion technique integrating image and point cloud data for rapid and accurate dimensional evaluation of precast bridge deck panels.

WORK EXPERIENCE

3D Modeler

2020-2022

Rendersoft Systems India Pvt. Ltd., Chennai, India

- Involved in modeling detailed BIM models of buildings based on architectural drawings using AutoCAD, 3Ds Max, and Blender.

CAD Trainer

CADD Centre Training Services, Chennai, India

2018-2020

- Conducted training for courses on 2D and 3D CAD modeling, and construction scheduling software. Also, I supervised college students on design projects to emphasize practical application of CAD tools in building design.

SCHOLARSHIPS

- Brain Korea Research Scholarship (BK21) by the Government of South Korea 2023-2025
- STEM Scholarship by Sungkyunkwan University 2023-2025

SKILLS

- **Programming:** MATLAB, Python, Rhinoceros scripting, and Dynamo visual programming for algorithm development
- **Computer Vision & AI:** Image and point cloud processing; synthetic image and point cloud training data generation through scan simulations
- **3D Modeling & Visualization:** Revit, 3Ds Max, and Blender for modeling and visualization tasks
- **Simulation & Robotics:** Webots software for robot-environment simulation
- **Reality Capture:** Handheld and terrestrial laser scanning
- **Aerial Imaging:** Drone operation (DJI Mavic series) for data collection and photogrammetry-based modeling

LANGUAGE PROFICIENCY

IELTS Academic Score: Band 7/9

2026

REFERENCES

- **Dr. Sung-Han Sim** (Professor)
School of Civil, Architectural Engineering and Landscape Architecture
Sungkyunkwan University
Relationship: Thesis Advisor
Email: ssim@skku.edu
- **Dr. Min-Koo Kim** (Associate Professor)
Department of Architectural Engineering
Chungbuk National University
Relationship: Thesis Co-advisor
Email: joekim@chungbuk.ac.kr